

Another instance of this class has access to the instance variables in any other instance of this class. You can use the instance identifier or the class name if it is declared as a “static” variable.

4.2.2 Cosmic Base Class

In Java, all classes are built on other classes. Ultimately, all classes in Java stem from one central “Cosmic Base Class” called Object. Even if you didn’t use the word “extends” in your class definition, you’re still always extending Object by default.

When you extend any “Base Class”, the new (derived) class has all the properties (instance variables) and methods of its parent, or Base Class. You can choose to modify or keep any method of the parent, or you can create methods that only apply to the child or “inherited” class.

The concept of extending a base class is called “Inheritance.” Inheritance is the second fundamental concept of Object-Oriented programming. Encapsulation is the first, Polymorphism is the third.

4.2.3 Relationships between classes

Classes can be related to each other in one of three alternative ways:

```
use
containment ( "has-a" )
inheritance ( "is-a" )
```

When one class sends messages to another class, we say it “uses” the class that receives its messages. When one class lives as an Instance Variable within another class, we say it is “Contained”, a “has-a” relationship. When one class inherits from another class, we say it is an “is-a” relationship.

Imagine that we have a class Order. Class Order needs to use the class Account, in order to check for credit status. Generally, if a method of class Order sends a message to an object of class Account, then Order uses Account. In other words, Order uses Account when Order calls methods of Account.

A method of Order: creates, receives or returns objects of class Account. The “Containment” relationship also known as the “Composition” relationship is a special case of the “use” relationship. In a Containment / Composition relationship, at least one method of one class actually contains an object of another class. In the use relationship, it calls methods of another